

Mesh Network Scheduling based on Cyber-Physical Sensitivity for Wireless Control Systems

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Abstract—Wireless control systems (WCSs) are gaining rapid development in industrial automation. Compared to the star topology, mesh networks offer greater compatibility for large-scale applications that require high reliability, scalability, and extended coverage. In WCSs, multiple control loops share the multi-hop mesh network, leading to non-negligible and long-span communication latency in critical flows, which can severely degrade the overall control performance. Additionally, the criticality of each control flow largely depends on the features of the physical plant dynamics and the mesh network configuration, which is essential to properly and exactly represent. Moreover, the online scheduling and reconfiguration for large-scale mesh network for WCSs also pose unique challenges. In this paper, we propose a mesh network scheduling mechanism based on cyber-physical sensitivity. Firstly, we model each control loop as a switched system to represent the impact of *arbitrary* and *fluctuating* communication latency. Second, we propose a novel online criticality indicator, *cyber-physical sensitivity (CP-Sensi)*, which accurately reflects the criticality of each control flow by synthesizing the switched model, run-time physical states, and network conditions. Finally, we design a *CP-Sensi-based scheduling mechanism* and an efficient piggyback-based *network reconfiguration protocol* tailored for mesh networks. Extensive studies with 12 control loops demonstrate that the proposed CP-Sensi and online mesh network scheduling achieve superior control performance compared to state-of-the-art approaches.

Index Terms—Wireless mesh network scheduling, cyber-physical systems, multi-loop control systems.

I. INTRODUCTION

Recent advancements in wireless communication and control technologies have made wireless control systems (WCSs) a promising solution for industrial automation, offering increased flexibility, and improved adaptability. A large-scale WCS, such as smart manufacturing [1] and the industrial production system [2], as shown in Fig. 1, consists of multiple field applications, comprising sensors, controllers, actuators, and physical plants. The interconnections among them are enabled by wireless mesh networks, through which multiple sensing/actuating flows deliver the states/control commands to their respective destinations. However, as the scale of the mesh network increases and more sensing and actuating devices are pouring into industrial fields, significant challenges emerge.

For large-scale WCSs with multiple loops over the mesh network, multi-hop transmissions lead to non-negligible communication latency. Consequently, the network scheduling mechanisms could lead to huge variances in latency of different control flows, resulting in different control performance,

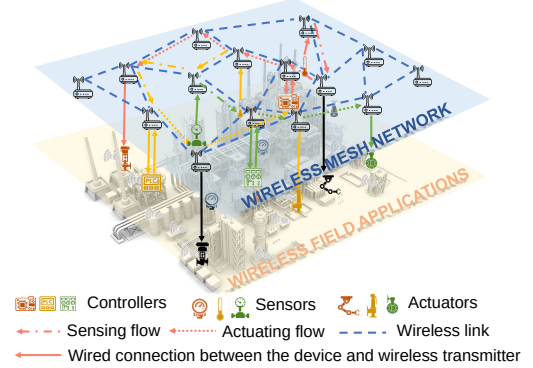


Fig. 1. Large-scale wireless control systems.

such as state errors represented by the difference between actual physical states and their reference values [3]. However, mainstream delay-aware network and control co-designs focus on a single type of delay, such as time-varying delays shorter [4], [5] or larger [6], [7] than the control period, or fixed delays [8], [9], which restricts their adaptability to large-scale mesh networks. Therefore, it is crucial to investigate the impacts of arbitrary and fluctuating delays.

In addition, the flow transmissions should be scheduled based on the runtime criticality to meet the diverse timing and control performance requirements of different loops. However, from the network perspective, the criticality of data flows is often determined by network utility functions (NUF), such as throughput [10] and delay [11], which focuses on communications performance without considering the dynamics of physical plants. From the network and control co-design perspective, criticality is often assessed using control performance metrics, such as the state quadratic [12], state error [13], root mean squared error (RMSE) [14], and Mahalanobis distance [6]. However, these metrics usually overlook the effects of fluctuating delays, network scheduling, and the interactions between flows in terms of resource occupation, transmission conflicts, and channel contentions. There is a lack of effective metrics to accurately quantify the criticality of transmitting each control flow, which depends on both physical plant dynamics and mesh network conditions, such as the network resource usage and the potential transmission conflicts.

While dynamic scheduling based on the criticality of data flows is essential, the lack of effective and reliable reconfiguration protocols for wireless mesh networks poses challenges. Unlike star networks, mesh networks face complexities due to an increase number of nodes and hops, making the dis-

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semination of reconfiguration signals and the overall network reconfiguration process non-trivial. As a result, the reconfiguration delay and energy consumption become critical factors.

In this work, we propose a mesh network scheduling mechanism for WCSs based on *cyber-physical sensitivity* (*CP-Sensi*). Our approach starts by modeling the WCS as a switched system under arbitrary and fluctuating delays. To evaluate the criticality of control flows over the shared mesh network, we propose a novel scheduling metric called *CP-Sensi*, which integrates the switched model, real-time physical states, and network conditions. Additionally, we develop a mesh network scheduling mechanism based on *CP-Sensi* and a reconfiguration protocol tailored for mesh networks. The primary contributions are four-fold:

- We introduce a switched model to analyze the impacts of arbitrary and fluctuating delays on control performance, characterizing system dynamics by capturing the distinct kinetics of subsystems and their switching trajectories.
- We define *cyber-physical sensitivity* (*CP-Sensi*) to represent the criticality of control flows, synthesizing the switched modeling under the arbitrary and fluctuating delays, the online physical states, and network conditions.
- We propose a priority-based scheduling mechanism based on *CP-Sensi* for wireless mesh networks. Additionally, we propose a *route design* for piggyback-based reconfiguration to improve reconfiguration efficiency, and conduct schedulability analysis.
- Comprehensive evaluations on 12 linear systems, along with comparisons to state-of-art methods, demonstrate the effectiveness of our approach across various utilization levels and network scales.

The rest of the paper is organized as follows. Sec. II reviews related work. Sec. III presents an overview of the framework. Sec. IV introduces the control system model under arbitrary and fluctuating latency. Sec. V defines cyber-physical sensitivity. Sec. VI presents the mesh network scheduling, along with the online reconfiguration protocol. Sec. VII describes the case studies in the context of 12 linear systems and a joint simulation, and we conclude the paper in Sec. VIII.

II. RELATED WORK

WCSs face significant challenges in meeting stringent timing and control requirements as the scale of mesh network increases, prompting extensive research efforts to enhance system performance. Existing works have focused on improving the timing performance, such as minimizing the average latency [15] or jitter [16]. However, the prevailing design principle is not tailored to WCS applications, as the control performance is not the design objective [17]. On the other hand, both offline [7], [18] and online [19]–[21] control-scheduling co-design approaches have gained popularity, which allocate network resources tailored for control performance. In multi-hop scenarios, Ghosh et al. [7] establish an aperiodic execution pattern that integrates scheduling and routing offline for multiple control loops. Klügel et al. [18] apply cross-layer optimization, covering sampling, transmission,

and feedback control. Chen et al. [19] propose joint uplink scheduling and channel allocation for WirelessHART. Jain et al. [20] develop adaptive, state-aware sampling rates to meet real-time constraints. However, these approaches represent each system as a discrete-time linear time-invariant (DT-LTI) model, which overlooks latencies shorter than the control period. Consequently, they are not suitable for large-scale mesh networks, where the arbitrary communication latencies significantly impact control performance. In contrast, Zhang et al. [4] and Luo et al. [5] address stability and stabilization under time-varying and delays shorter than control period, respectively, but they do not model the impact of fine-grained delays on the dynamic control performance.

To address diverse real-time requirements of multiple industrial applications, several network utility functions have been proposed for optimizing network performance. Gu et al. [10] propose a co-design for rate control and flow scheduling to maximize network utility. Neely et al. [11] develop a delay-based NUF to ensure deterministic worst-case delay guarantees. Additionally, criticality-aware scheduling [6], [14], [22] addresses varying real-time and control performance needs by cyber-physical states. Mager et al. [6] allocate bandwidth to agents with the highest communication needs using Mahalanobis distance. Wang et al. [14] dynamically prioritize control packets based on the RMSE of previous measurements. Roy et al. [22] allocate multi-QoS resources to ensure each loop meets its settling time requirements. However, these criticality metrics do not equate to scheduling sensitivity in wireless mesh networks. Specifically, relying on a single control metric for mesh network scheduling overlooks the interconnection among control performance of multiple loops, resource usage, and potential transmission conflicts.

Another major challenge in dynamic mesh network scheduling is achieving energy-efficient and timely online reconfiguration. Broadcasting is used by the network reconfiguration in [19], [20], [23]. The Low-Power Wireless Bus (LWB) protocol [24] supports rapid data communication and network reconfiguration based on *Glossy* [25]. However, due to limited capacity and energy consumption, broadcasting imposes the significant overheads [26]. In contrast, piggyback embeds reconfiguration messages within transmitted packets, offering an energy-efficient alternative [27], [28]. However, the dissemination latency of piggyback highly depends on the original schedule and its feasibility. Neither approach sufficiently addresses the need to improve reconfiguration efficiency for wireless mesh networks.

III. WIRELESS CONTROL SYSTEM ARCHITECTURE

In this section, we present the modelings of WCSs and mesh network communication, and describe the framework of our proposed *CP-Sensi*-based mesh network scheduling.

A. Modeling of the WCSs

A WCS is schematically depicted in Fig. 2. The continuous-time linear and time-variant model of each plant is

$$\dot{x}_i(t) = A_i x_i(t) + B_i u_i(t - d_i(t)), \quad (1)$$

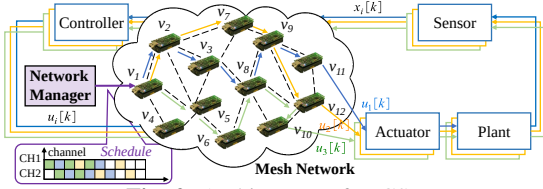


Fig. 2. Architecture of WCSs.

where $i \in \{1, \dots, N\}$ denotes the control loop index, t is the time variable, $A_i \in \mathbb{R}^{n \times n}$ and $B_i \in \mathbb{R}^{n \times m}$ are the system matrices. $x_i(t) \in \mathbb{R}^n$ and $u_i(t) \in \mathbb{R}^m$ represent the physical states and the control commands, respectively. $d_i(t)$ is the delay suffered by $u_i(t)$ during mesh network communication.

N control loops share the limited resources of the mesh network. Each loop comprises a continuous-time plant, sensors, actuators, and a discrete-time controller. Similar to the system setups in [29], [30], time-driven sensors send $x_i[k]$ to the controllers over the wireless network at $t = kT_i$, incurring a sensor-to-controller delay $d_{i,s}(t)$. At $t = kT_i + d_{i,s}(t)$, the event-driven controllers generate control commands $u_i[k]$ and send them to the actuators via the wireless mesh network, experiencing a controller-to-actuator delay $d_{i,a}(t)$. The event-driven actuators, with a zero-order-hold, execute the control commands upon arrival at $t = kT_i + d_i(t)$, where the end-to-end delay is $d_i(t) = d_{i,s}(t) + d_{i,a}(t)$.

B. Wireless Mesh Network

Our network model applies multi-channel Time Division Multiple Access (TDMA) protocol, which has the same salient features as WirelessHART [31] and WIA [32]. Each node is equipped with a half-duplex radio. Time is divided into slots with each slot assigned to only one deterministic link. Multiple channels allows per-time slot channel hopping. To avoid co-channel interference, we do not consider the spatial reuse of channels. Please note that the contents before Sec. VI-C assumes that routing is arbitrarily given. We will explore the degree of freedom and potential in route design in Sec. VI-C2. A centralized Network Manager (NM) is responsible for the overall network management includes scheduling, channel allocation, and routing, while controllers and the NM are connected to the gateway via wired links. We assume the mesh network provides reliable communications but incurs scheduling latency due to multi-hop transmissions [33]–[36].

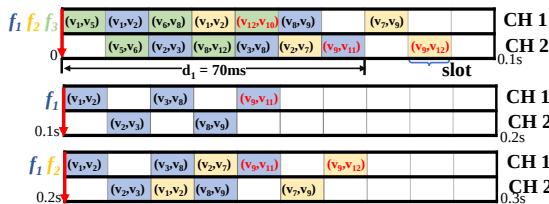


Fig. 3. TDMA slot. (Different colors in the slots represent different loops. Following the topology in Fig. 2, $\mathcal{R}_1 = [(v_1, v_2), (v_2, v_3), (v_3, v_8), (v_8, v_9), (v_9, v_{11})]$, $\mathcal{R}_2 = [(v_1, v_2), (v_2, v_7), (v_7, v_9), (v_9, v_{12})]$, $\mathcal{R}_3 = [(v_1, v_5), (v_5, v_6), (v_6, v_8), (v_8, v_{12}), (v_{12}, v_{10})]$, the final hop in $\mathcal{R}_i$ is marked in red to indicate the latency of f_i .)

The wireless mesh network is modeled as an undirected graph $G(V, E, \Lambda)$, where $V = \{v_1, \dots, v_{|V|}\}$ represents the set

of nodes, $e_{s,d} = (v_s, v_d) \in E$ denotes a wireless link from v_s to v_d , and Λ indicates the number of channels. Each time slot has a duration of T_{slot} . As the impacts of latency on the uplink (sensors to controllers) and downlink (controllers to actuators) differ significantly [37], we focus primarily on downlink scheduling [22], [38] and assume that the uplink can use any existing scheduling approaches, implementing asymmetric uplink/downlink scheduling in this work [39], [40]. The control commands generated by each controller are transmitted as a data flow $f_i \in \mathcal{F}$ from the controller (source) to actuator (destination), where $\mathcal{F}$ represents the set of flows. Each f_i is characterized by a 4-tuple $\langle T_i, D_i, P_i, \mathcal{R}_i \rangle$, corresponding to the period T_i , the implicit deadline $D_i = T_i$, the priority P_i , and the route $\mathcal{R}_i$ from the controller to actuator. Control periods T_i are integral multiples of the smallest control period T_s . $\mathcal{R}_i = [\mathcal{R}_i[1], \dots, \mathcal{R}_i[H_i]]$ denotes the route of f_i , where $\mathcal{R}_i[h] \in E$ and H_i is the number of hops for $\mathcal{R}_i$. Each f_i requires H_i time slots to deliver a packet through $\mathcal{R}_i$.

For instance, in Fig. 3, $f_1 \triangleq \langle 100 \text{ ms}, 100 \text{ ms}, 2, \mathcal{R}_1 \rangle$, $f_2 \triangleq \langle 200 \text{ ms}, 200 \text{ ms}, 3, \mathcal{R}_2 \rangle$, and $f_3 \triangleq \langle 300 \text{ ms}, 300 \text{ ms}, 1, \mathcal{R}_3 \rangle$, $\mathcal{R}_i$ is specified in the caption, $T_{slot} = 10 \text{ ms}$. Under the fixed-priority scheduling, although $H_1 = H_3 = 5$, f_3 experiences 50 ms latency, f_1 50 ms and 70 ms, and f_2 70 ms and 90 ms with different transmission conflicts among flows, demonstrating the variations in delays among flows and within the same flow. These delays directly impact the control performance, making it crucial to dynamically adjust priorities by the criticality of different flows.

C. CP-Sensi-Based Scheduling Framework

This section introduces the CP-Sensi-based mesh network scheduling framework, which enhances control performance dynamically for large-scale WCSs.

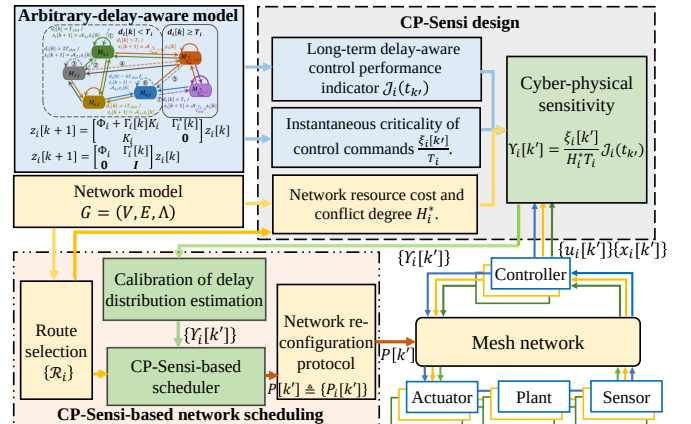


Fig. 4. Framework of the CP-Sensi-based network scheduling.

As illustrated in Fig. 4, the physical plant is first modeled as a switched system with arbitrary and fluctuating delays. Based on this arbitrary-delay-aware physical model and the network model $G(V, E, \Lambda)$, the **CP-Sensi design** module constructs cyber-physical sensitivity, which combines both long-term and short-term physical performance with network conditions to indicate the criticality of different control flows. The **CP-Sensi-based network scheduling** module then manages

network scheduling and reconfiguration based on *CP-Sensi*. This module includes a CP-Sensi-based network scheduler for optimizing control performance, route design to reduce reconfiguration latency and improve reliability, and iterative calibration of delay distribution estimation.

For deployment, we offline construct the arbitrary-delay-aware model in the *CP-Sensi design* module, and the route design in the *CP-Sensi-based network scheduling* module in NM. During the online stage, at each coordination period (super-frame length) T_c with time index k' , NM calculates CP-Sensi $\{\Upsilon_i[k']\}$ based on $\{u_i[k']\}$ and $\{x_i[k']\}$, and sends these to the *CP-Sensi-based scheduling* module to calibrate the delay distribution estimation. Through the reconfiguration protocol, the NM disseminates the priority set $P[k'] \triangleq \{P_1[k'], \dots, P_N[k']\}$, proposed by the CP-Sensi-based scheduler. Once all wireless nodes receive $P[k']$, they implicitly generate a global, conflict-free schedule by $\langle T_i, D_i, P_i, R_i \rangle$. Then, at each slot of T_{slot} , the network transmits f_i with $u_i[k]$ according to this schedule.

IV. SYSTEM MODELING UNDER ARBITRARY AND FLUCTUATING END-TO-END DELAYS

As a key variable in the interaction between physical and network systems, the end-to-end delay is both arbitrary and fluctuating. Accurately modeling how these time-varying delays influence physical states is essential for effective control and scheduling design, though challenging to achieve. Switched systems are widely employed in fault-tolerant systems and communication networks [41]. Leveraging this approach, we model the dynamics under each fixed delay as a distinct subsystem, with fluctuations characterized by the switching trajectory among these subsystems.

Unlike models that impose constraints on the delay range and distribution [4], [5], we consider any $d_i[k] > 0$ at the k -th control period. The discrete-time model of the WCS is represented by:

$$x_i[k+1] = \Phi_i x_i[k] + \Gamma_i^*[k] u_i^*[k] + \Gamma_i[k] u_i[k], \quad (2)$$

where

$$\Phi_i = e^{A_i T_i}, \Gamma_i^*[k] = \int_0^{d_i[k]} e^{A_i t} B_i dt, \Gamma_i[k] = \int_{d_i[k]}^{T_i} e^{A_i t} B_i dt, \quad (3)$$

$x_i[k]$ denotes the physical state at the sampling instant t_k where $t_{k+1} - t_k = T_i$, $u_i[k]$ is calculated within the k -th period T_i by the controller, and $u^*[k]$ is the last control commands stored at the actuator [42]. The control law is:

$$u_i[k] = K_i x_i[k], \quad (4)$$

where K_i is the static feedback gain.

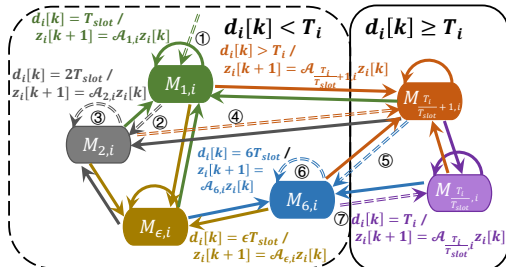


Fig. 5. A FSM of modeling under fluctuating arbitrary delays.

To model the system dynamics in Eqs. (2) and (4) as a linear switched system, we propose a new augmented state vector $z_i[k] = [x_i^T[k], u_i^{*T}[k]]^T$ to denote the state of the switched system at t_k . The closed-loop dynamics are defined as follows:

- For $0 < d_i[k] \leq T_i$, since the duration of a time slot T_{slot} is the minimal unit of delay, the ϵ -th subsystem model, corresponding to delay $d_i[k] = \epsilon T_{slot}$ with $\epsilon \in \{1, 2, \dots, \frac{T_i}{T_{slot}}\}$, can be denoted as:

$$z_i[k+1] = \mathcal{A}_{\epsilon,i} z_i[k] = \begin{bmatrix} \Phi_i + \Gamma_i[k] K_i & \Gamma_i^*[k] \\ K_i & \mathbf{0} \end{bmatrix} z_i[k]. \quad (5)$$

- For $d_i[k] > T_i$, which implies $u_i[k]$ generated at time t_k misses its deadline, and $u_i[k]$ is discarded (denoted as *miss*), and the subsystem model is uniformly given by:

$$z_i[k+1] = \mathcal{A}_{\epsilon,i} z_i[k] = \begin{bmatrix} \Phi_i & \Gamma_i' \\ \mathbf{0} & \mathbf{I} \end{bmatrix} z_i[k], \quad (6)$$

where $\epsilon = \frac{T_i}{T_{slot}} + 1$, $\Gamma_i' = \int_0^{T_i} e^{A_i t} B_i dt$ as the last control commands $u_i^*[k]$ are actuated at k -th period.

- In the subsequent $(k+1)$ -th period, the model is recursively calculated by Eq. (5) or (6), depending on $d_i[k+1]$.

Φ_i , $\Gamma_i[k]$, and $\Gamma_i^*[k]$ are shown in Eq. (3). $\mathbf{0}$ and $\mathbf{I}$ denote zero and identity matrices of appropriate dimensions. In the following sections, $z_i[k]$ is equivalent to $z_i(t_k)$.

Fig. 5 presents a finite state machine (FSM) that describes the WCS model under fluctuating and arbitrary delays. Each state represents the current system model as one of the subsystems defined in Eqs. (5) and (6) during the k -th control period, and the arrows between states denote the transitions, labeled with ‘guard/action’. The ‘guard’ specifies the transition condition, i.e., $d_i[k]$, while the ‘action’ describes the response of switching. This switched model and switching trajectories encapsulate the online dynamics of the plant in the presence of fluctuating and arbitrary delays. For instance, when the fluctuating delay sequence follows $T_{slot} \rightarrow 2T_{slot} \rightarrow 2T_{slot} \rightarrow miss \rightarrow 6T_{slot} \rightarrow 6T_{slot} \rightarrow T_i$ shown by double-line arrows at time index k , the system states at time index $k+7$ becomes $z_i[k+7] = \mathcal{A}_{1,i} \mathcal{A}_{2,i}^2 \mathcal{A}_{\frac{T_i}{T_{slot}}+1,i} \mathcal{A}_{6,i}^2 \mathcal{A}_{\frac{T_i}{T_{slot}}+1,i} z_i[k]$.

Based on the maximal eigenvalue $\lambda_{\max}(\mathcal{A}_{\epsilon,i})$, as the linear system in (3) becomes unstable when the delays exceed a delay margin [43], we can divide $\frac{T_i}{T_{slot}} + 1$ subsystems into stable set $\mathcal{M}_i^s = \{\{M_{1,i}, \dots, M_{r_i,i}\} | \lambda_{\max}(\mathcal{A}_{\epsilon,i}) < 1\}$ and unstable set $\mathcal{M}_i^{us} = \{\{M_{r_i+1,i}, \dots, M_{\frac{T_i}{T_{slot}}+1,i}\} | \lambda_{\max}(\mathcal{A}_{\epsilon,i}) \geq 1\}$, r_i is the number of stable subsystems. In Sec. V, we develop CP-Sensi with consideration of the stability of these subsystems.

V. CYBER-PHYSICAL SENSITIVITY OF WCSS

In this section, we introduce CP-Sensi for the mesh network scheduling tailored to WCSSs. Unlike the traditional indicators of flow criticality, such as state quadratic [12], we define flow sensitivity as a combination of the effects of arbitrary and fluctuating delays and the network conditions on the control performance at run-time, indicating the scheduling importance (priority) of f_i . First, we present a novel long-term control performance indicator based on the arbitrary-delay-aware model in Sec. IV. Furthermore, we discuss cyber-

physical sensitivity to characterize the criticality of f_i , which is used by our mesh network scheduling mechanism in Sec. VI.

A. A Novel Delay-Aware Control Performance Indicator

In the WCS studies, the linear quadratic (LQ) cost is the most commonly applied run-time control performance metric [12], [14], [23]. At time step k with the horizon T_c ,

$$\mathcal{J}_i^{\text{LQ}}[k] = \sum_{\tau=0}^{T_c/T_i} z_i[k+\tau]^\top Q_i z_i[k+\tau], \quad (7)$$

where Q_i is a semi-definite matrix. However, it cannot adequately predict control performance under arbitrary and fluctuating delays. At time step k , future states $z_i[k+\tau]$ for $\tau \in \{1, \dots, \frac{T_c}{T_i}\}$ are predicted, which are strongly intertwined with mesh network scheduling. Since these predicted states are unavailable until the schedule is determined, they cannot, in turn, be used to guide network scheduling decisions. For multi-hop mesh networks, the performance indicator requires further interaction with network scheduling. Additionally, with a coordination period T_c , the total prediction steps $\frac{T_c}{T_i}$ for heterogeneous loops vary unfairly. Therefore, we focus on addressing two key questions to establish a novel control performance indicator.

- **Q1:** How to predict the control performance for the arbitrary-delay-aware model Eqs. (5) and (6)?
- **Q2:** How to predict the control performance for multiple control loops with different control periods T_i ?

As discussed in Sec. IV, the arbitrary-delay-aware modeling identifies both stable and unstable subsystems. For stable subsystems, the convergence rates of state trajectories provide detailed characterization of performance differences. However, for unstable subsystems, where non-converging states grow exponentially, these cannot be assessed along with those of the stable subsystems. Motivated by this, for **Q1**, we design separate performance metrics for each type of subsystem. Specifically, the indicator includes statistical metrics for the states of the stable subsystems and quantitative measures for the number of unstable subsystems. For **Q2**, We standardize the prediction step across loops by obtaining a continuous evolution bound function (EBF) for each stable subsystem.

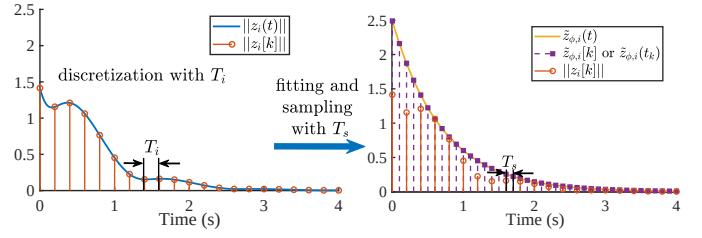
1) *The Evolution Bound Function (EBF):* we first introduce EBF to describe the evolution of r_i stable subsystems, which is the prerequisite for solving **Q1** and **Q2**.

Definition 1. [44] The DT-LTI system in Eq. (5) or (6) is exponentially stable if there exists constants $\alpha_{d,i} \in (0, 1)$ and $c_{d,i} > 0$ that $\forall z_i[k] \in \mathbb{R}^n, \tau \in \mathbb{N}^+$,

$$\|z_i[k+\tau]\| \leq c_{d,i} \cdot \alpha_{d,i}^\tau \|z_i[k]\|. \quad (8)$$

For a given d_i , once we derive $\alpha_{d,i}$ and $c_{d,i}$, we can characterize the state evolution of a stable subsystem. To efficiently solve **Q2**, we transform Eq. (8) into a continuous format by Eq. (9), which we denote the evolution bound function (EBF) of the ϕ -th subsystem ($\phi = d_i/T_{\text{slot}}$) as $\tilde{z}_{\phi,i}$, $\phi \in \{1, \dots, r_i\}$. This allows all loops to be sampled at a uniform step length,

$$\|z_i(\lfloor \frac{t_k + \tau}{T_i} \rfloor \cdot T_i)\| \leq \tilde{z}_{\phi,i}(t_k + \tau) \triangleq \tilde{c}_{d,i} \cdot e^{-\tilde{\alpha}_{d,i}\tau} \|z_i[k]\|. \quad (9)$$



(a) Sampling the continuous state by T_i . (b) Sampling the EBF with T_s .
Fig. 6. Fitting and sampling of the continuous EBF $\tilde{z}_{\phi,i}(t)$.

For any stable d_i and $\|z_i[k]\|$, the continuous $\tilde{z}_{\phi,i}$ can be fitted using $\tilde{\alpha}_{d,i}$ and $\tilde{c}_{d,i}$ to minimize errors between $\tilde{z}_{\phi,i}(t_k + \tau)$ and the discrete $\|z_i(\lfloor \frac{t_k + \tau}{T_i} \rfloor \cdot T_i)\|$, as shown in Fig. 6. Notably, the model is time-invariant, allowing us to set $t_k = 0$ and calculate $\tilde{\alpha}_{d,i}$ and $\tilde{c}_{d,i}$ offline using Eq. (10).

$$\begin{aligned} \min_{\tilde{\alpha}_{d,i}, \tilde{c}_{d,i}} \lim_{\Delta \rightarrow 0} \sum_{k=0}^{\infty} \tilde{c}_{d,i} \cdot e^{-\tilde{\alpha}_{d,i}k\Delta} \|z_i[0]\| - \|z_i(\lfloor \frac{k\Delta}{T_i} \rfloor \cdot T_i)\|, \\ \text{s.t. } \tilde{c}_{d,i} \cdot e^{-\tilde{\alpha}_{d,i}k\Delta} \|z_i[0]\| - \|z_i(\lfloor \frac{k\Delta}{T_i} \rfloor \cdot T_i)\| \geq 0, \quad (10) \\ 1 > \tilde{\alpha}_{d,i} > 0, \tilde{c}_{d,i} > 0. \end{aligned}$$

During the run-time operation, we can determine $\tilde{z}_{\phi,i}(t)$, $\forall t \geq t_k$ by the stored $(\tilde{\alpha}_{d,i}, \tilde{c}_{d,i})$ for the current $z_i[k]$.

2) *Delay-Aware Control Performance Indicator:* Based on the EBFs in Eq. (9), we can characterize the statistic of $\tilde{\mathbf{z}}_i(t) = \{\tilde{z}_{\phi,i}(t)\}$, $\phi \in \{1, \dots, r_i\}$ at any time $t \geq t_k$, for stable subsystems of loop i by their expectation and variance.

$$\mathbb{E}(\tilde{\mathbf{z}}_i(t)) = \sum_{\phi=1}^{r_i} w_{\phi,i} \cdot \tilde{z}_{\phi,i}(t), \quad (11)$$

$$\text{Var}(\tilde{\mathbf{z}}_i(t)) = \sum_{\phi=1}^{r_i} w_{\phi,i} \cdot (\tilde{z}_{\phi,i}(t) - \mathbb{E}(\tilde{\mathbf{z}}_i(t)))^2, \quad (12)$$

where $w_{\phi,i}$ is the weight of ϕ -th stable subsystem of loop i , which is initially identical to $\frac{1}{r_i}$ and further calibrated in Sec. VI-A. At each time $t > t_k$, $\mathbb{E}(\tilde{\mathbf{z}}_i(t))$ denotes the expectation of physical states of r_i stable subsystems, while $\text{Var}(\tilde{\mathbf{z}}_i(t))$ reflects the robustness of the control loop under delays within $\{T_{\text{slot}}, \dots, r_i T_{\text{slot}}\}$. For unstable subsystems, we use the quantity $|\mathcal{M}_i^{us}|$ to represent the resilience of the control system to longer delays. To address **Q1**, the control performance indicator Eq. (13) for loop i includes two parts: the numerical statistics of the EBF values for stable subsystems and the quantity of unstable subsystems.

$J_i(t) = (\mathbb{E}(\tilde{\mathbf{z}}_i(t)) \mathcal{V}(\text{Var}(\tilde{\mathbf{z}}_i(t))) \mathbb{E}(\tilde{\mathbf{z}}_i(t)))^{\frac{1}{2}} + \beta |\mathcal{M}_i^{us}|$, (13) where β is the weight the unstable term. $\mathcal{V}(\cdot)$ is a function of $\text{Var}(\tilde{\mathbf{z}}_i(t))$, which is given by:

$$\mathcal{V}(\text{Var}(\tilde{\mathbf{z}}_i(t))) = e^{\text{Var}(\tilde{\mathbf{z}}_i(t))^p}, \quad (14)$$

where $0 < p < 1$ is the coefficient exponent of $\text{Var}(\tilde{\mathbf{z}}_i(t))$. Unlike the LQ cost $\mathcal{J}_i^{\text{LQ}}[k]$, by incorporating variance into the performance indicator, Eq. (13) effectively represents the delay sensitivity of loop i . Adjusting p in $\mathcal{V}(\cdot)$ can alter the weight between average control performance (expectation) and latency sensitivity (variance). It is interesting that the value of $J_i(t)$ is dominated by the second term $\beta |\mathcal{M}_i^{us}|$ when loop i nearly converges to the equilibrium ($z_i[k] \rightarrow \mathbf{0}$),

allowing a more effective and intuitive characterization of static properties, such as stability.

Finally, as shown in Fig. 6b, we replace the step length of loop i from T_i to the uniform T_s on the EBF within the prediction interval T_c to address **Q2**. The delay-aware control performance indicator at time t then becomes

$$\mathcal{J}_i(t) = \sum_{\tau=0}^{T_c/T_s} J_i(t + \tau \cdot T_s). \quad (15)$$

Remark V.1. Eq. (15) with Eq. (13) is a generalized model of Eq. (7). Without loss of generality, it can be used in both switched and time-invariant systems. For a stable DT-LTI system, it is obvious that the $\mathbb{E}(\tilde{z}_i[k]) = z_i[k]$, $\mathcal{V}(\text{Var}(\tilde{z}_i[k])) = 1$, $\mathcal{M}_i^{us} = \emptyset$, and then $J_i(t_k) = z_i^\top(t_k)z_i(t_k)$.

B. Cyber-Physical Sensitivity

Although $\mathcal{J}_i(t)$ can represent control performance over a long-term T_c , it does not fully capture scheduling criticality, which should include multi-time-granularity control performance and mesh network conditions. Specifically, the *instantaneous importance of control flows*, *network resource cost*, and *transmission conflicts*, i.e., a transmission of this flow and that of a higher priority flow involve a common node [45], also impact the criticality of f_i . Therefore, we propose cyber-physical sensitivity (CP-Sensi) for loop i as:

$$\Upsilon_i[k] = \mathcal{F}(\mathcal{J}_i(t_k), \xi_i[k], H_i^*). \quad (16)$$

The CP-Sensi consists of two parts. The first part is the multi-time-granularity control performance indicator, which includes the finite-horizon control performance indicator $\mathcal{J}_i(t_k)$ and the instantaneous criticality of control commands $\xi_i[k]$ at time t_k . The second part is the resource usage and conflict degree indicator H_i^* . However, for different WCSs, $\mathcal{F}(\cdot)$ may vary to better represent scheduling sensitivity. In this paper, we provide a heuristic but effective form of $\mathcal{F}(\cdot)$, which demonstrates its superiority in Sec. VII.

$$\Upsilon_i[k] = \frac{\xi_i[k]}{H_i^* T_i} \mathcal{J}_i(t_k). \quad (17)$$

The choices of $\xi_i[k]$ and H_i^* are detailed as follows.

1) *Network Resource Cost and Conflict Degree H_i^** : When multiple loops share limited network resource, transmission conflicts increase end-to-end delay. Specifically, if two flows conflict by sharing a common node, the lower-priority flow may be delayed. As shown in Fig. 3, during $0 \sim 0.1$ s, $\mathcal{R}_1[1] = (v_1, v_2)$ of f_1 must be scheduled at slot 2, while $\mathcal{R}_2[3] = (v_7, v_9)$ must delay by one slot as the radio of v_9 is occupied by $\mathcal{R}_1[5]$ in slot 7. Since the degree of conflict correlates with the number of common nodes with other flows, we define the conflict degree as the sum of the counts that each node in $\mathcal{R}_i$ is covered by all flows, i.e.,

$$H_i^* = \sum_{v_q \in \mathcal{R}_i} \sum_{l=1}^N \sum_{h=1}^{H_l} \mathbf{1}_{\mathcal{R}_l[h]=(\cdot, v_q) \vee \mathcal{R}_i[h]=(v_q, \cdot)}, \quad (18)$$

which also reflects network resource cost as H_i^* is related to H_i . Here, $\mathbf{1}_{\text{condition}}$ is an indicator function that is 1 if the condition is true and 0 otherwise. $\frac{\mathcal{J}_i(t_k)}{H_i^*}$ thus represents the criticality per unit of network resource and conflict costs.

2) *Instantaneous Criticality of Control Commands $\xi_i[k]$* : We define the criticality time frame $\xi_i[k]$ of $u_i[k]$ to describe its instantaneous criticality. In Eq. (2), the control commands $u_i[k]$ and $u_i^*[k]$ drive the evolution of $x_i[k]$, represented as the weighted evolution by $u_i[k]$ and $u_i^*[k]$, as shown on the right-hand side of Eq. (19). Given a certain value of ρ , $\xi_i[k]$ can be derived. A larger $\xi_i[k]$ indicates that the state evolution leans more on $u_i[k]$ than $u_i^*[k]$, making $\xi_i[k]$ a suitable indicator of the criticality of $u_i[k]$. In Eq. (17), we further use $\frac{\xi_i[k]}{T_i}$ instead of $\xi_i[k]$ to represent the instantaneous criticality tailored for heterogeneous flows.

$$\underbrace{\int_{T_i - \xi_i[k]}^{T_i} e^{A_i t} B_i dt u_i[k] + \int_0^{T_i - \xi_i[k]} e^{A_i t} B_i dt u_i^*[k]}_{\text{The evolution of } x_i[k] \text{ with input-delay of } d_i[k] = T_i - \xi_i[k]} = (1 - \rho) \underbrace{\int_0^{T_i} e^{A_i t} B_i dt u_i[k]}_{\text{with ideal inputs}} + \rho \underbrace{\int_0^{T_i} e^{A_i t} B_i dt u_i^*[k]}_{\text{with delayed inputs}} \quad (19)$$

VI. MESH NETWORK SCHEDULING

This section presents the CP-Sensi-based scheduling and reconfiguration protocol tailored for mesh networks. We first present the online calibration of the delay distribution estimation in Sec. VI-A. The scheduling mechanism is summarized in Sec. VI-B. Sec. VI-C proposes our online reconfiguration protocol. Sec. VI-D provides the schedulability analysis.

A. The Calibration of the Delay Distribution Estimation

Eqs. (11) and (12) initially set the distribution of delay candidates equally, i.e., $w_{\phi, i} = \frac{1}{r_i}$, which should be calibrated by the actual resulted latency distribution by the CP-Sensi-based network scheduling. The calibration can be performed using fixed point iteration on $w_{\phi, i}$, based on the interdependency between the cost function $\mathcal{J}_i(t_{k'})$, which depends on the latency distribution, and CP-Sensi-based scheduling based on $\mathcal{J}_i(t_{k'})$, which contributes to the actual latency distribution. As shown in Alg. 1, $w_{\phi, i}[k']$ is initialized as $w_{\phi, i}^{[0]}[k'] = \frac{1}{r_i}$ (Line 3). In the ψ -th iteration, the priority-based scheduler generates the slot schedule in descending order of $\Upsilon_i^{[\psi]}[k']$ (Line 5). Based on the schedule, the delay candidates $\{d_i[k]\}$, $\kappa \in \{\frac{k'T_c}{T_i}, \frac{k'T_c}{T_i} + 1, \dots, \frac{(k'+1)T_c}{T_i}\}$ that f_i suffers during $[k'T_c, (k'+1)T_c]$ can be determined. If $d_i[k] = \phi T_{slot}$, $w_{\phi, i}^{[\psi]}[k']$ is incremented by a fixed step Δ_i (Lines 6 - 12). Finally, $w_{\phi, i}^{[\psi+1]}[k']$ is calculated by normalizing $w_{\phi, i}^{[\psi]}[k']$ (Line 13). After Ψ rounds of calibration, $w_{\phi, i}[k'] = w_{\phi, i}^{[\Psi]}[k']$ and $\Upsilon_i[k'] = \Upsilon_i^{[\Psi]}[k']$ (Line 16). The iteration step Δ_i should be chosen carefully to prevent oscillation and inefficiency. The impact of Ψ is evaluated in Sec. VII-D. The computational complexity of schedule generation within $[k'T_c, (k'+1)T_c]$ by the preemptive fixed-priority scheduler [45] is $O(\Lambda \cdot T_c \cdot \sum_{i=1}^N H_i \cdot T_i)$ where Λ is the number of channels. Therefore, Alg. 1 has a computational complexity of $O(\Psi \cdot T_c \cdot \sum_{i=1}^N (\Lambda \cdot H_i \cdot T_i))$.

B. CP-Sensi-based Mesh Network Scheduling

We apply a priority-based scheduler based on CP-Sensi. The priorities ranged from 1 to N in each coordination period T_c

Algorithm 1 Calibration of the delay distribution estimation

```

1: Input:  $\langle T_i, D_i, P_i[k'], \mathcal{R}_i \rangle, r_i, \Delta_i, T_{slot}, T_s, T_c, \Lambda$ , and  $\Psi$ .
2: Output:  $\{w_{\phi,i}[k']\}, \{\Upsilon_i[k']\}, \phi \in \{1, \dots, r_i\}, i \in \{1, \dots, N\}$ .
3: Initialize:  $w_{\phi,i}^{[0]}[k'] = \frac{1}{r_i}$ , and  $\Upsilon_i^{[0]}[k'], \phi \in \{1, \dots, r_i\}$ ;
4: for  $\psi = 0$  to  $\Psi - 1$  do
5:   calculate  $\{\mathcal{J}_i^{[\psi]}(t_{k'})\}$ , generate the schedule of  $[k'T_c, (k' + 1)T_c]$ .
6:   for  $i = 1$  to  $N$  do
7:     get  $\{d_i[\kappa]\}, \kappa \in \{\frac{k'T_c}{T_i}, \frac{k'T_c}{T_i} + 1, \dots, \frac{(k' + 1)T_c}{T_i}\}$ ;
8:     for  $\kappa = \frac{k'T_c}{T_i}$  to  $\frac{(k' + 1)T_c}{T_i}$  do
9:       if  $d_i[\kappa] = \phi T_{slot}$  then
10:         $w_{\phi,i}^{[\psi]}[k'] \leftarrow w_{\phi,i}^{[\psi]}[k'] + \Delta_i$ ;
11:       end if
12:     end for
13:      $w_{\phi,i}^{[\psi+1]}[k'] \leftarrow \frac{w_{\phi,i}^{[\psi]}[k']}{\sum_{\phi=1}^{r_i} w_{\phi,i}^{[\psi]}[k']}$ ;
14:   end for
15: end for
16:  $\{w_{\phi,i}[k']\} \leftarrow \{w_{\phi,i}^{[\Psi]}[k']\}, \{\Upsilon_i[k']\} \leftarrow \{\Upsilon_i^{[\Psi]}[k']\}$ .
17: return  $\{w_{\phi,i}[k']\}, \{\Upsilon_i[k']\}, \phi \in \{1, \dots, r_i\}, i \in \{1, \dots, N\}$ .

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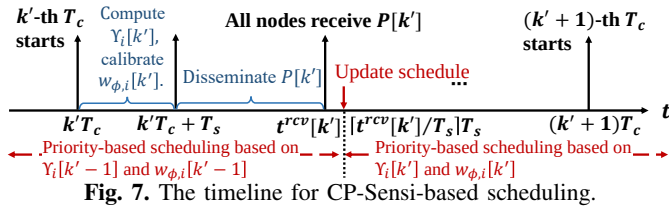


Fig. 7. The timeline for CP-Sensi-based scheduling.

are assigned in descending order of $\Upsilon_i[k']$. As shown in Fig. 7, the calculation of CP-Sensi, calibration of delay distribution estimation, and the network scheduler cause overhead at each T_c indexed by k' . To accommodate this, we reserve a fixed computation interval of one T_s , which is sufficient and further discussed in Sec. VII-D. Consequently, during $[k'T_c, k'T_c + T_s]$ of the k' -th T_c , the network still follows the previous schedule from the $(k' - 1)$ -th T_c . Starting from $k'T_c + T_s$, the NM disseminates the priorities $P[k'] \triangleq \{P_i[k']\}$ by the previous schedule via our reconfiguration protocol. At $t = \lceil \frac{t^{rcv}[k']}{T_s} \rceil T_s$, each node generates the new global schedule by the preemptive fixed-priority scheduler [45]. For the new schedule, instances of all flows f_i , where $i \in \{1, \dots, N\}$, following $P_i[k' - 1]$ that are still active (released earlier but not completed) are assigned higher priorities than to that of instances following $P_i[k']$ that are released after $t = \lceil \frac{t^{rcv}[k']}{T_s} \rceil T_s$. The global schedule repeats every T_c until $t = \lceil \frac{t^{rcv}[k'+1]}{T_s} \rceil T_s$, and the network transmits flows with $u_i[k]$ according to it at each slot.

At $t = k'T_c$, each node follows the schedule by the previous $P_i[k' - 1]$ and $T_i, i \in \{1, 2, \dots, N\}$. Then it computes the number of slots, denoted as ϖ , at which all nodes have received $P[k']$ at least once from $t = k'T_c + T_s$ during run-time. Therefore, the time to achieve global convergence during reconfiguration in the k' -th coordination period is $t^{cv}[k'] = k'T_c + \varpi T_{slot}$. Finally, at $t = \lceil \frac{t^{rcv}[k']}{T_s} \rceil T_s$, all nodes start to transmit according to the new schedule based on $P[k']$, achieving global convergence. The time cost ϖT_{slot} for global convergence can be derived by searching the schedule with $P_i[k' - 1]$ to find the latest node, among those covered by all N flows, that receives the packet with updated $P[k']$. As it is closely related to the dynamic priority $P[k' - 1]$ stored in each

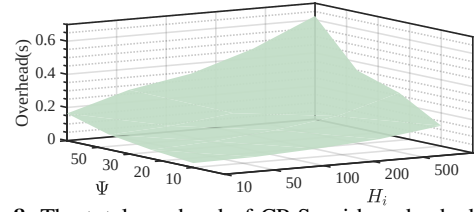


Fig. 8. The total overhead of CP-Sensi-based scheduling.

node, we can offline estimate the upper bound of the global converging time. Under any priority assignment, the time for receiving $P[k]$ by all nodes is upper-bounded by the highest worst-case response time, i.e., $\max_{P_i, i \in \{1, \dots, N\}} \max_i R_i$, where R_i is the worst-case end-to-end delay for f_i by the response time analysis in Sec. VI-D.

In Fig. 8, we present the overhead of the CP-Sensi-based scheduler under different Ψ and H_i which reflects the network scale. The overall computational complexity is $O(\Psi \cdot T_c \cdot \sum_{i=1}^N (\Lambda \cdot H_i \cdot T_i) + N \cdot \log N)$ which is contributed by Alg. 1 and the fixed-priority scheduler with $O(N \cdot \log N)$. The scalability is demonstrated as the overhead remains under 0.6 s for a 500-hop network.

Remark VI.1. Empirically, the range of an optimal T_c can be $[T_s + \max_{P_i, i \in \{1, \dots, N\}} \max_i R_i, \min_i \varsigma_i]$, where $T_s + \max_{P_i, i \in \{1, \dots, N\}} \max_i R_i$ is the upper bound of the network reconfiguration delay, ς_i the time constant of loop i , since an ideal T_c should be chosen to both balance the reconfiguration and timely response to physical disturbances.

C. Network Reconfiguration Protocol

One of the key challenges for dynamic scheduling for mesh networks is to design the effective reconfiguration protocol.

1) *Network Reconfiguration Based on Piggyback:* Broadcast facilitates timely dissemination but at a heavy energy cost, particularly for frequent reconfiguration in mesh networks [26]. To address this, we employ *piggyback* to disseminate reconfiguration messages. Due to limited payload, e.g., the maximum payload of WirelessHART is 102 B [46], directly disseminating the schedule for large-scale networks is impossible. Therefore, the NM appends the updated priorities $P[k'] \triangleq \{P_i[k']\}$ to the control commands $u_i[k]$, where $i \in \{1, \dots, N\}$. The data field of this packet becomes $[u_i[k], P[k']]$, based on which the global schedule is generated implicitly.

2) *Routing Design for Effective Reconfiguration:* The above contents work for arbitrarily given routes. However, routing designs have a significant impact on both the reconfiguration and flow delays [45]. In this section, we explore the degree of freedom and potential in route design heuristically to reduce both the reconfiguration and flow delays. As discussed in Fig. 3, the worst-case latency of the reconfiguration is 250 ms if it happens at $t = 0.2$ s. In another instance, if $\mathcal{R}_3 = [(v_1, v_5), (v_5, v_6), (v_6, v_8), (v_8, v_{10})]$, $\mathcal{R}_2 = [\mathcal{R}_3, (v_{10}, v_{12})]$, and $\mathcal{R}_1 = [\mathcal{R}_3, (v_{10}, v_{12}), (v_{12}, v_{11})]$, the reconfiguration latency fixes at 60 ms, as all nodes of $\mathcal{R}_2$ and $\mathcal{R}_3$ are contained in $\mathcal{R}_1$. In this case, the workload overly occupies nodes within $\mathcal{R}_3$ and the latency of all flows increases. From the example, we can find three key factors for reducing both reconfiguration

and flow delays: (1) the length of routes, (2) the number of nodes covered by different routes, and (3) the balance of node utilization, which inspire our following route design.

Accordingly, we first define the partial route $\mathcal{R}_i^m$ as a part of $\mathcal{R}_i$ from the actuator node to the latest relay node v_m , which is already established as the latest intermediate result of $\mathcal{R}_i$ during the route construction by Alg. 2. ϱ_i^m is defined as the resource usage of $\mathcal{R}_i^m$ by the period and the length of route,

$$\varrho_i^m = \frac{H_i^m}{T_i}, \quad (20)$$

where H_i^m denotes the number of hops of $\mathcal{R}_i^m$. As the reconfiguration latency is relevant to the coverage counts of every node in the routes, we define *common coverage benefit* ε_i^m of $\mathcal{R}_i^m$, which represents the sum of the coverage counts of each $v_p \in \mathcal{R}_i^m$ by N partial routes established simultaneously, i.e., $\mathcal{R}_j^{n_j}$, for $j \in \{1, \dots, N\}$ in Alg. 2. Greater ε_i^m leans to shorter dissemination latency. Nodes with less coverage count might dominate the dissemination latency. Meanwhile, excessive reuse of a single specific node increases transmission conflicts and delays. Thus, we tend to increase the coverage counts while balance among multiple flows. We define the *load balancing cost* ϑ_i^m , which shows the maximal difference in the number of transmissions between v_m and its neighbor v_q .

$$\varepsilon_i^m = \sum_{v_p \in \mathcal{R}_i^m} \sum_{j=1}^N \sum_{h=1}^{H_j^{n_j}} \frac{1}{T_j} \mathbf{1}_{\mathcal{R}_j^{n_j}[h]=(\cdot, v_p)} \vee \mathcal{R}_j^{n_j}[h]=(v_p, \cdot), \quad (21)$$

$$\vartheta_i^m = \max_{v_q} \sum_{j=1}^N \sum_{h=1}^{H_j^{n_j}} \frac{1}{T_j} |\mathbf{1}_{\mathcal{R}_j^{n_j}[h]=(\cdot, v_m)} \vee \mathcal{R}_j^{n_j}[h]=(v_m, \cdot) - \mathbf{1}_{\mathcal{R}_j^{n_j}[h]=(\cdot, v_q)} \vee \mathcal{R}_j^{n_j}[h]=(v_q, \cdot)}|. \quad (22)$$

Finally, we define the total benefit Φ_i^m of $\mathcal{R}_i^m$ as

$$\Phi_i^m = \eta_\varepsilon \cdot \varepsilon_i^m - \eta_\vartheta \cdot \vartheta_i^m - \eta_\varrho \cdot \varrho_i^m, \quad (23)$$

where η_ε , η_ϑ , and η_ϱ are customized nonnegative coefficients. According to the total benefit Φ_i^m , we can select the relay nodes considering the above three factors.

We first label each node as L_ω^s , with its minimal hop ω to the access point (AP) and the index s in the all nodes that are minimal ω hop away from AP, demoted as L_ω , $\omega \in \{0, \dots, \Omega\}$, $s \in \{1, \dots, S_\omega\}$. To reduce the searching space and network usage, we only allow one node to route to the others with the same or smaller ω . Alg. 2 heuristically selects the relay nodes in L_ω for all routes simultaneously, from $\omega = \Omega$ to 0, as shown in Fig. 9. For each loop i , given $v_{p,i} \in L_{\omega+1}$, it determines $v_{q,i} \in L_\omega$ of $(v_{p,i}, v_{q,i})$ by maximizing Φ_i^q , such that $\mathcal{R}_i^q = [\mathcal{R}_i^p, (v_{p,i}, v_{q,i})]$ is derived (Lines 5-13). After N loops determine their $\mathcal{R}_i^q$, we extend the partial routes within the same L_ω . Given $v_{q,i} \in L_\omega$, it iteratively determines the relay nodes $v_{m,i}, v_{m',i} \in L_\omega$ by maximizing Φ_i^m and $\Phi_i^{m'}$ (Lines 14-26). The last node $v_{m,i} \in L_\omega$ is set as the start point $(v_{p,i})$ of the next layer, i.e., $\omega - 1$ (Line 27). Alg. 2 returns each $\mathcal{R}_i$ by reversing $\mathcal{R}_i^{\text{AP}}$ (Line 29).

D. Schedulability Analysis

We analysis schedulability for both the certain schedule and schedule update process, respectively.

Algorithm 2 Routing design for effective reconfiguration.

```

1: Input:  $G(V, E, \Lambda)$ ,  $T_i$ ,  $\eta_\varepsilon$ ,  $\eta_\vartheta$ , and  $\eta_\varrho$ .
2: Output:  $\mathcal{R}_i$ ,  $i \in \{1, \dots, N\}$ .
3: Initialize:  $v_{p,i} \leftarrow$  the actuator node of  $f_i$ ,  $\Phi_i^p \leftarrow 0$ ,  $\mathcal{R}_i^p \leftarrow \emptyset$ .
4: for  $\omega$  from  $\Omega$  to 0 do
5:   for loop  $i$  in descending order of  $\Phi_i^p$  do
6:     for  $s \in \{1, \dots, S_\omega\}$  do
7:       if  $(v_{p,i}, L_\omega^s) \in E$  then
8:         calculate  $\Phi_i^{L_\omega^s}$  for  $[\mathcal{R}_i^p, (v_{p,i}, L_\omega^s)]$ .
9:       end if
10:    end for
11:     $v_{q,i} \leftarrow \operatorname{argmax}_{s \in \{1, \dots, S_\omega\}} \Phi_i^{L_\omega^s}$ ;
12:     $\mathcal{R}_i^q \leftarrow [\mathcal{R}_i^p, (v_{p,i}, v_{q,i})]$ .
13:  end for
14:  for loop  $i$  in descending order of  $\Phi_i^q$  do
15:    Initialize:  $v_{m,i} \leftarrow v_{q,i}$ ,  $v_{m',i} \leftarrow v_{m,i}$ ;
16:    while  $\Phi_i^m \geq \Phi_i^{m'}$  do
17:       $v_{m',i} \leftarrow v_{m,i}$ .
18:      for  $s \in \{1, \dots, S_\omega\}$  do
19:        if  $(v_{m',i}, L_\omega^s) \in E$  then
20:          calculate  $\Phi_i^{L_\omega^s}$  for  $[\mathcal{R}_i^{m'}, (v_{m',i}, L_\omega^s)]$ .
21:        end if
22:      end for
23:       $v_{m,i} \leftarrow \operatorname{argmax}_{s \in \{1, \dots, S_\omega\}} \Phi_i^{L_\omega^s}$ , s.t.  $\Phi_i^{L_\omega^s} \geq \Phi_i^{m'}$ .
24:       $\mathcal{R}_i^m \leftarrow [\mathcal{R}_i^{m'}, (v_{m',i}, v_{m,i})]$ .
25:    end while
26:  end for
27:   $v_{p,i} \leftarrow v_{m,i}$ ,  $\mathcal{R}_i^p \leftarrow \mathcal{R}_i^m$ .
28: end for
29: return each  $\mathcal{R}_i = \text{reverse}(\mathcal{R}_i^{\text{AP}})$ ,  $i \in \{1, \dots, N\}$ .

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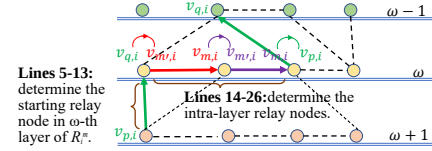


Fig. 9. Diagram of the routing design.

Schedulability of certain schedule. Schedulability under fixed-priority scheduling can be obtained using response time analysis (RTA) [45], which we apply to analyze schedulability under both $P[k' - 1]$ and $P[k']$ offline.

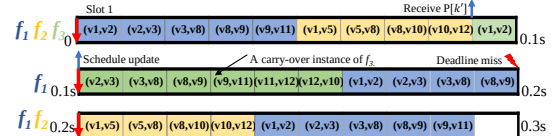


Fig. 10. An illustrative example of the priority change.

Schedulability of the schedule update process. When the schedule updates at $t = \lceil \frac{t^{rcv}[k']}{T_s} \rceil T_s$ within the superframe duration, there may be flows that have not yet finished transmissions which runs under $P[k' - 1]$ and are called carry-over instances. Fig. 10 shows a 3-flow and single-channel example, with $f_1 \triangleq \langle 100 \text{ ms}, 100 \text{ ms}, 1, \mathcal{R}_1 \rangle$, $f_2 \triangleq \langle 200 \text{ ms}, 200 \text{ ms}, 2, \mathcal{R}_2 \rangle$, and $f_3 \triangleq \langle 300 \text{ ms}, 300 \text{ ms}, 3, \mathcal{R}_3 \rangle$, where $\mathcal{R}_1$, $\mathcal{R}_2$, and $\mathcal{R}_3$ are shown in the schedule table. All nodes receive the updated $P[k'] \triangleq [P_1[k'], P_2[k'], P_3[k']] = [2, 1, 3]$ at $t = 0.09 \text{ s}$ and the schedule updates when $t = 0.1 \text{ s}$, with the slot length of 10 ms. Although the flows are schedulable under both $P[k' - 1]$ and $P[k']$, the first instance of f_3 which occupies slots 10-16 has the highest priority and preempts the second instance of f_1 , which misses its deadline when $t = 0.2 \text{ s}$. Therefore, given that the system is schedulable under $P[k' - 1]$ and $P[k']$, respectively, we

analyze schedulability of the schedule update process using workload analysis [47], [48]. For analysis purpose, we assume the flows in $\mathcal{F}$ are ordered by decreasing priority, and denote $\mathcal{F}_i[k'] \triangleq \{f_1, \dots, f_i\}$ as the subset of i highest priority flows under $P[k']$.

Since the system is schedulable under $P[k' - 1]$ and a flow's deadline is no more than its period, there is at most one carry-over instance per flow. Besides the channel contention, transmission conflicts with higher priority flows than f_i causes additional delays to f_i . The total demand $\mathcal{D}_i(\sigma)$ by at most N carry-over instances and the new instances of i highest priority flows during $[\lceil \frac{t^{rcv}[k']}{T_s} \rceil T_s, \lceil \frac{t^{rcv}[k']}{T_s} \rceil T_s + \sigma]$ can be divided into $\mathcal{D}_i^{ch}(\sigma)$ due to channel contention and $\mathcal{D}_i^{con}(\sigma)$ due to transmission conflicts.

We first analyze the total demands of carry-over and new instances of $\mathcal{F}_i[k']$ caused by channel contention, i.e.,

$$\mathcal{D}_i^{ch}(\sigma) = \mathcal{D}_{co}^{ch}(\sigma) + \mathcal{D}_{new,i}^{ch}(\sigma), \quad (24)$$

where $\mathcal{D}_{co}^{ch}(\sigma)$ is the demand of at most N carry-over transmissions, $\mathcal{D}_{new,i}^{ch}(\sigma)$ the demand of the new instances of i highest priority flows following $P[k']$. For a carry-over instance of $f_j, j \in \{1, \dots, N\}$, the maximum remaining execution time during $[\lceil \frac{t^{rcv}[k']}{T_s} \rceil T_s, \lceil \frac{t^{rcv}[k']}{T_s} \rceil T_s + \sigma]$ has two options: 1) $H_j T_{slot}$, if $\sigma \geq D_j$; 2) $\min\{\sigma, H_j T_{slot}\}$, if $\sigma < D_j$. Then, the total demand $\mathcal{D}_{co,i}^{ch}(\sigma)$ of N carry-over instances by channel contention in any length σ as

$$\mathcal{D}_{co}^{ch}(\sigma) = \frac{1}{\Lambda} \sum_{j \in \mathcal{F}_N[k' - 1]} \min\{\sigma, H_j T_{slot}\}, \quad (25)$$

where Λ is the number of channels. For the **new** instances following $P[k']$, the demand $\mathcal{D}_{new,i}^{ch}$ of i highest priority flows by channel contention in any interval length σ is

$$\mathcal{D}_{new,i}^{ch}(\sigma) = \frac{1}{\Lambda} \sum_{f_i \in \mathcal{F}_i[k']} \max\{0, \lfloor \frac{\sigma - D_i + T_i}{T_i} \rfloor H_i T_{slot}\} \quad (26)$$

where $\lfloor \frac{\sigma - D_i + T_i}{T_i} \rfloor$ is the maximum number of instances of f_i during $[\lceil \frac{t^{rcv}[k']}{T_s} \rceil T_s, \lceil \frac{t^{rcv}[k']}{T_s} \rceil T_s + \sigma]$.

For the network demands of carry-over and new instances of flows within $\mathcal{F}_i[k']$ caused by transmission conflicts, i.e.,

$$\mathcal{D}_i^{con}(\sigma) = \mathcal{D}_{co}^{con}(\sigma) + \mathcal{D}_{new,i}^{con}(\sigma), \quad (27)$$

For carry-over instances, as there is at most one carry-over instance per flow, the maximum number of conflicts between f_j and $f_q \in \mathcal{F}_{j-1}[k' - 1]$ is one. Hence, for at most N carry-over instances, the $\mathcal{D}_{co}^{con}(\sigma)$ can be denoted as:

$$\mathcal{D}_{co}^{con}(\sigma) = \sum_{j \in \mathcal{F}_N[k' - 1]} \sum_{f_q \in \mathcal{F}_{j-1}[k' - 1]} \Theta(j, q), \quad (28)$$

where $\Theta(j, q)$ is an upper bound of the delay that f_j can experience from an instance of $f_q \in \mathcal{F}_{j-1}[k' - 1]$ [45, Lemma 1]. For the new instances of f_i and $f_q \in \mathcal{F}_{i-1}[k']$, each instance of f_i can be conflicted by f_q at most $\lceil \frac{\sigma}{T_q} \rceil$ times. Each instance of f_i can be conflicted by a carry-over instance of $f_p \in \mathcal{F}_N[k' - 1] \setminus f_i$ at most $\lceil \frac{\sigma}{T_p} \rceil$ times. For the transmission conflicts, the demand $\mathcal{D}_{new,i}^{con}(\sigma)$ of flow f_i , and that of i highest priority flows can be denoted as Eqs. (29) and (30), respectively.

$$\mathcal{D}_{new,i}^{con}(\sigma) = \sum_{f_q \in \mathcal{F}_{i-1}[k']} \left\lceil \frac{\sigma}{T_q} \right\rceil \Theta(i, q) + \sum_{f_p \in \mathcal{F}_N[k' - 1] \setminus f_i} \left\lceil \frac{\sigma}{T_p} \right\rceil \Theta(i, p), \quad (29)$$

$$\mathcal{D}_{new,i}^{con}(\sigma) = \sum_{f_i \in \mathcal{F}_i[k']} \mathcal{D}_{new,i}^{con}(\sigma). \quad (30)$$

Lemma VI.1. [48, Lemma 2] *The worst-case workload of at most N carry-over instances and i highest priority flows under $P[k']$ during $P[k' - 1] \rightarrow P[k']$ can be denoted as*

$$W_i(v) = \min_{\sigma \in [0, v]} (\mathcal{D}_i^{ch}(\sigma) + \mathcal{D}_i^{con}(\sigma) + (v - \sigma)). \quad (31)$$

where $\mathcal{D}_i^{ch}(\sigma)$ and $\mathcal{D}_i^{con}(\sigma)$ are defined in Eqs. (24) and (27), respectively.

Theorem VI.2 states the schedulable condition.

Theorem VI.2. *The system is schedulable during a priority change from $P[k' - 1]$ to $P[k']$ if*

- 1) *the system is schedulable by fixed-priority scheduling under both priorities $P[k' - 1]$ and $P[k']$,*
- 2) $\forall i \in \{1, \dots, N\}, \exists \sigma \in [0, D_i] : H_i T_{slot} + \mathfrak{d}_{new,i}^{con}(D_i) + \mathcal{D}_{i-1}^{ch}(\sigma) + \mathcal{D}_{i-1}^{con}(\sigma) \leq \sigma.$

Proof. We prove that condition 2) holds. Without loss of generality, we can assume that for the schedulability of flow f_i following $P[k']$, all flows $f_1, \dots, f_{i-1}$ are schedulable. In this case, the schedulability condition of f_i can be denoted as

$$H_i T_{slot} + \mathfrak{d}_{new,i}^{con}(D_i) + W_{i-1}(D_i) \leq D_i \quad (32)$$

where $\mathfrak{d}_{new,i}^{con}(D_i)$ is the network demand of f_i by transmission conflicts, as shown in Eq. (29). $W_{i-1}(D_i)$ is the total workload of N carry-over instances and $i-1$ highest priority flows under $P[k']$ in Lemma VI.1.

$$H_i T_{slot} + \mathfrak{d}_{new,i}^{con}(D_i) + \min_{\sigma \in [0, D_i]} \mathcal{D}_{i-1}^{con}(\sigma) + \mathcal{D}_{i-1}^{ch}(\sigma) + (D_i - \sigma) \leq D_i,$$

$$H_i T_{slot} + \mathfrak{d}_{new,i}^{con}(D_i) + \min_{\sigma \in [0, D_i]} \mathcal{D}_{i-1}^{con}(\sigma) + \mathcal{D}_{i-1}^{ch}(\sigma) - \sigma \leq 0,$$

$$\exists \sigma \in [0, D_i], H_i T_{slot} + \mathfrak{d}_{new,i}^{con}(D_i) + \mathcal{D}_{i-1}^{con}(\sigma) + \mathcal{D}_{i-1}^{ch}(\sigma) \leq \sigma. \quad (33)$$

Hence, the schedulability condition for all flows is given by $\forall i \in \{1, \dots, N\}, \exists \sigma \in [0, D_i] : H_i T_{slot} + \mathfrak{d}_{new,i}^{con}(D_i) + \mathcal{D}_{i-1}^{ch}(\sigma) + \mathcal{D}_{i-1}^{con}(\sigma) \leq \sigma.$ $\square$

For instance, ten flows are defined by f_1 to f_{10} in Sec. VII-A with 5 channels. T_i (D_i) and H_i for each flow (control loop) are shown in Table. I and II, respectively. The derivation of different utilization U is described in Sec. VII-A. Fig. 11 shows the acceptance ratio for our analysis when priority change from $P[k' - 1]$ to $P[k']$, RTA for both $P[k' - 1]$ and $P[k']$ without priority change, RTA for $P[k' - 1]$, RTA for $P[k']$, and the actual simulation with priority change. For each point, 10,000 test cases are randomly selected among $10!^2 - 10!$ possible priority changes. During each priority change, the other specifications of each flow, such as period and deadline, remain unchanged, and we change the utilization in the same way as Sec. VII-B by reducing the number of available time slots. When the network resources are sufficient, i.e., utilization $U < 0.4$, if $P[k' - 1]$ and $P[k']$ are both schedulable according to RTA (the orange curve), the transition between $P[k' - 1]$ and $P[k']$ has a high probability of passing our schedulability analysis. However, the gap between our analysis and the simulation results increases when $U \geq 0.4$, due to the pessimistic estimation of demands caused by transmission conflicts to flows with higher priority.

Remark VI.2. *Based on the condition 1) of Theorem VI.2, we firstly use RTA to derive the set of all schedulable priority settings without priority changes, denoted as $\mathbf{P}^{sche}$.*

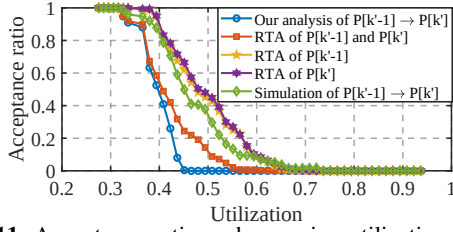


Fig. 11. Acceptance ratio under varying utilization levels.

Subsequently, according to condition 2), we apply workload analysis in Eq. (33) to derive the set of all schedulable priority transitions, denoted as $\{[P^{old}, P^{new}]\}^{sche}$, where $P^{old}, P^{new} \in \mathcal{P}^{sche}$. At each run-time step k' , the system is schedulable if $[P[k' - 1], P[k']] \in \{[P^{old}, P^{new}]\}^{sche}$.

Remark VI.3. Although RTA and workload analysis in our schedulability analysis are both pseudopolynomial, whether it can be adopted as an admission test should be based on the scale of real-time applications [48]. Therefore, currently, we analyze the schedulability offline to find all schedulable cases and their switches for a given task set, as testing the schedulability online, especially under all potential schedules, is relatively computationally expensive. In the future, we will study co-optimizing the control performance and schedulability analysis online by candidate mode selection, and heuristics with probabilistic schedulability analysis, etc.

VII. EVALUATION

In this section, we present an evaluation of the mesh network scheduling and the functionalities of CP-Sensi in multi-loop WCSs scenarios. Our investigation encompasses performance assessment across 12 heterogeneous control loops.

A. Experimental Setups

1) *Physical Plants*: Consider 12 independent 2-state continuous-time, e.g., DC-motor models [22], [49] in Eq. (1) with matrices in Tab. I. The control performance is evaluated in terms of mean absolute error $MAE = \frac{1}{W} \sum_{k=0}^W |x_i[k] - x_{ref,i}[k]|$, where W is the total number of samples within the 30s simulation interval. The initial and reference states of each loop are $x_i[0] = [10 \ 10]^T$ and $x_{ref,i}[k] = [0 \ 0]^T$, respectively. Without loss of generality, we set the origin of $\mathbb{R}^n$ as the equilibrium point, since any equilibrium point can be shifted to the origin via a transformation of variables [44].

TABLE I: System matrices of 12 control loops.

Loop	$\{A_i, B_i, K_i\}$	$T_i(s)$	Loop	$\{A_i, B_i, K_i\}$	$T_i(s)$
1	$A = \begin{bmatrix} -1 & 1 \\ -1.8 & 1.8 \end{bmatrix}$	0.4	4	$A = \begin{bmatrix} 0 & 1 \\ 0 & 0.6 \end{bmatrix}$	0.4
2	$B = [0 \ 2]^T$	0.2	5	$B = [0 \ 1]^T$	0.3
3	$K = -[2 \ 1]$	0.1	6	$K = -[7 \ 4]$	0.1
7	$A = \begin{bmatrix} 0 & 1 \\ 1 & 0.8 \end{bmatrix}$	0.3	10	$A = \begin{bmatrix} -1 & 1 \\ -1.8 & 1.8 \end{bmatrix}$	0.1
8	$B = [0 \ 1]^T$	0.2	11	$B = [0 \ 1]^T$	0.2
9	$K = -[5 \ 3]$	0.1	12	$K = -[5 \ 8]$	0.2

To validate the reliability of the scheduling against sudden interference to the physical plants, we add pulse physical disturbances by a Dirac pulse to the plant state x_i of loops 1 ~ 12 at $t = 12, 24$ s; loops 2, 6, and 7 at $t = 7$ s; loops 2, 3, 6, 8, 10, and 12 at $t = 9, 20$ s; loops 1 and 9 at $t = 15$ s.

2) *Mesh Network*: The evaluation results from Secs. VII-B to VII-D are based on our devised real-time wireless network simulator employing MATLAB/Simulink, which enables real-time scheduling of IEEE 802.15.4 mesh networks on the fly and accurately reflects the impact of real-time scheduling on run-time physical states. In Sec. VII-E, we simulate a realistic IEEE 802.15.4 network using the TOSSIM simulator [50], and the corresponding setups will be introduced.

Each link is configured to allow only one transmission attempt. Slot length is 10 ms, i.e. $T_{slot} = 10$ ms. The default network scale consists of 24 nodes and 2 channels. The topology and original routes are shown in Fig. 12. We explore more scales of the network later in Sec. VII-B3. Setting $\eta_e = 0.6$, $\eta_\varepsilon = 0.8$, and $\eta_\vartheta = 0.1$, the new routes established by Alg. 2 are shown in Tab. II. Since changing the control periods causes unfair comparisons of control performance under different utilization, we adjust the utilization U of N control loops among their available network resources by reserving the last ζ slots per $T_s = 100$ ms across all channels for other uses, i.e., $U = \sum_{i=1}^N \frac{U_i}{N * \Lambda}$, where $U_i = \frac{H_i T_{slot}}{T_i}$, and $\Lambda = \frac{T_s - \zeta T_{slot}}{T_s}$ is the reservation coefficient. For $\zeta = 0, 1, 2, 3$, U is 0.85, 0.94, 1.06, and 1.21, respectively.

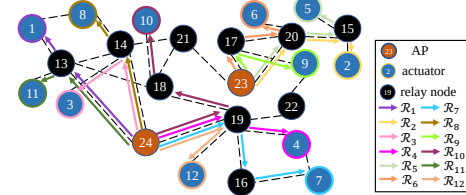


Fig. 12. The network topology and the original routes.

TABLE II: The new routes $\mathcal{R}_i$ by the routing algorithm Alg. 2.

$\mathcal{R}_1 = [(v_{24}, v_{13}), (v_{13}, v_1)]$	$\mathcal{R}_7 = [(v_{24}, v_{19}), (v_{19}, v_4), (v_4, v_7)]$
$\mathcal{R}_2 = [(v_{23}, v_{20}), (v_{20}, v_{15}), (v_{15}, v_2)]$	$\mathcal{R}_8 = [(v_{24}, v_{13}), (v_{13}, v_1), (v_1, v_8)]$
$\mathcal{R}_3 = [(v_{23}, v_{14}), (v_{14}, v_3)]$	$\mathcal{R}_9 = [(v_{23}, v_{17}), (v_{17}, v_9)]$
$\mathcal{R}_4 = [(v_{24}, v_{19}), (v_{19}, v_4)]$	$\mathcal{R}_{10} = [(v_{23}, v_{21}), (v_{21}, v_{18}), (v_{18}, v_{10})]$
$\mathcal{R}_5 = [(v_{23}, v_{20}), (v_{20}, v_{15}), (v_{15}, v_5)]$	$\mathcal{R}_{11} = [(v_{24}, v_{13}), (v_{13}, v_{11})]$
$\mathcal{R}_6 = [(v_{23}, v_{20}), (v_{20}, v_6)]$	$\mathcal{R}_{12} = [(v_{24}, v_{19}), (v_{19}, v_{12})]$

3) *Comparison Groups*: We compare the performance under the state-of-the-art approaches.¹

- The proposed network scheduling based on CP-Sensi.
- Dynamic Priority Adjustment: $P_i[k]$ adjusts every T_c by:
 - Finite-horizon state quadratic (LQ) [18]
 - Current state error (SE) [23]
 - RMSE of past measurements (RMSE) [14]
- Average Latency First (ALF): ALF adapts the priorities every T_c by sorting the average delay in previous T_c .
- Latency Minimization (LM) [15]: LM selects the optimal priorities, which minimize the average delay of all flows.
- Jitter Minimization (JM) [16]: JM selects the optimal priorities, which minimize the average jitter of all flows.
- Dynamic Period Assignment (DPA) [51]: Periods are adapted every T_c to minimize MAE. Flows follow RM.
- Period Selection (PS) [52]: PS selects optimal control periods offline to minimize MAE. Flows follow RM.
- Real-time Scheduling Algorithm: earliest deadline first (EDF), rate monotonic (RM), least laxity first (LLF).

¹For fairness, all dynamic scheduling strategies employ our proposed reconfiguration protocol in Sec. VI-C.

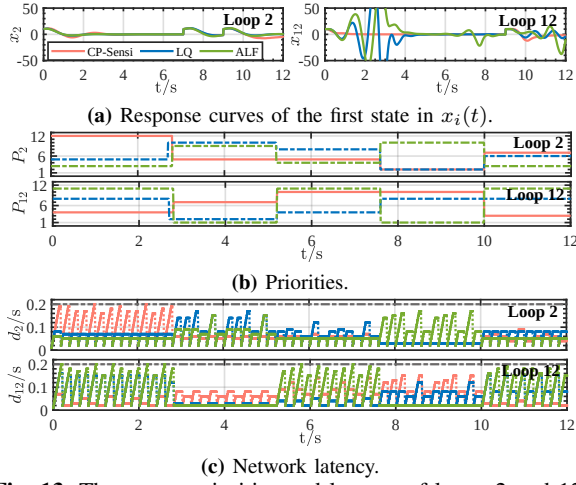


Fig. 13. The states, priorities and latency of loops 2 and 12.

B. Performance Comparison

For CP-Sensi, $p = 0.2$ in Eq. (14) and $\beta = 0.2$ in Eq. (17). The coordination period T_c is set to 2.4 s and the maximum iteration round is $\Psi = 10$. We evaluate T_c and Ψ in Secs. VII-B4 and VII-D, respectively.

1) *Response Curve*: We take Case 2 ($U = 0.94$) as an example. The response curves of the first state in $x_i(t)$, priority settings, and network latency for two typical loops (loops 2 and 12) are shown in Fig. 13. We also present results of LQ and ALF which are representative scheduling approaches based on physical and network states, respectively. Compared to CP-Sensi, LQ adjusts scheduling effectively according to the performance predictions in Eq. (7) when only a few loops are disturbed and eager for control command updates, e.g., $P_2 = 2$ at $t = 7.6$ s. However, when most loops are eager for updates of control commands, LQ cannot effectively adjust their scheduling priorities. For Loop 12 at $t = 0$ s and $t = 9$ s, LQ fails to capture the impact of fluctuating and arbitrary delays, nor does it consider network factors. In contrast, CP-Sensi can better utilize resources by representing the dynamic, fine-grained network requirements to the control system based on comprehensive factors, i.e., the arbitrary-delay-aware modeling, the network usage and conflicts degree, and the instantaneous criticality of $u_i[k]$. For example, CP-Sensi sets $P_2 = 12$ at $t = 0$ s, the physical state $x_2(t)$ is similar to that of LQ whose $P_2 = 5$. Additionally, CP-Sensi raises P_{12} since it is highly sensitive to delays, with 11 out of the total 21 subsystems in Eqs. (5) and (6) being unstable. For ALF, the priorities and latency fluctuate with a period of T_c , as it does not consider the dynamic evolution of physical states. As a result, the control performance deteriorates.

2) *Statistical Control Performance*: We analyze the statistical control performance under different scheduling schemes and utilization levels. Fig. 14 shows the average MAE obtained from 20 rounds of simulation across various initial physical states. Each data point represents the overall MAEs for 12 loops over $0 \sim 30$ s. Missing data points for EDF, RM, and LLF reflect poor control performance in certain loops, demonstrating the importance of control-aware scheduling.

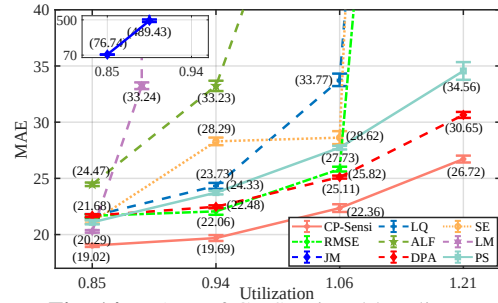


Fig. 14. MAEs of CP-Sensi and baselines.

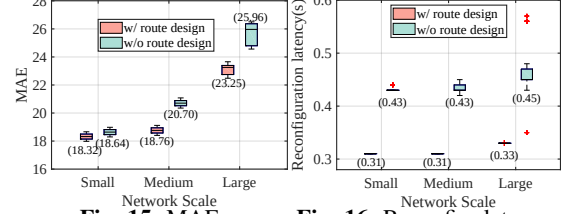


Fig. 15. MAE.

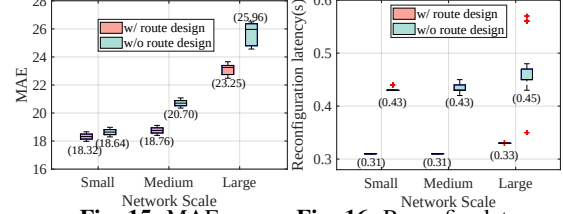


Fig. 16. Reconfig. latency.

Compared to other scheduling approaches, CP-Sensi significantly enhances control performance. Under $U = 0.85$, LQ, SE, and RMSE allow most loops to transmit control packets on time. Although RMSE is calculated from the past measurements, it achieves the best performance among the three metrics because it represents the transient response using the known states. The fluctuating performance of SE stems from limited ability to indicate dynamic control performance accurately, particularly when the transient state is near equilibrium. The MAE of LQ is up to 51.0% larger than that of CP-Sensi, highlighting that effective control performance prediction requires interaction with network scheduling. Similarly, the MAE of DPA is up to 14.7% higher than that of CP-Sensi, as DPA should meet hard real-time demands, leading to performance degradation. PS, which statically selects periods without accounting for physical disturbances, is outperformed by CP-Sensi up to 29.3%. Approaches like ALF, LM, and JM, which aim to optimize latency or jitter without considering control performance, struggle to manage limited network resources effectively under dynamic control scenarios.

3) *Reconfiguration*: We present three network scales: small, medium (default), and large. The small- and large-scale networks consist of 17 and 38 nodes, respectively. All configurations use 4 channels. Figs. 15 and 16 display the control performance and reconfiguration latency with (w/) and without (w/o) route design, respectively. Across all network scales, the reconfiguration latency with route design remains under 0.4 s, averaging 120 ms less than that without route design, as nodes in other flows with $T_i = 0.4$ s (loops 1 and 4) are contained in the other flows whose $T_i \leq 0.3$ s by Alg. 2. In the small-scale network, they perform similarly since flows encounter fewer delays. In the large-scale network, route design improves control performance by 11.6%.

4) T_c : As illustrated in Fig. 17, as T_c increases, the MAE first decreases within the range of $T_c = 0.4$ to 2.4 s. Due to reconfiguration latency shown in Fig. 16, overly frequent reconfigurations prevent the system from fully utilizing pri-

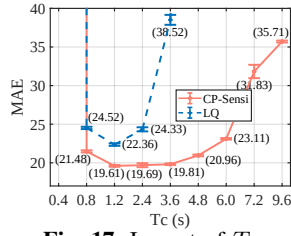


Fig. 17. Impact of T_c .

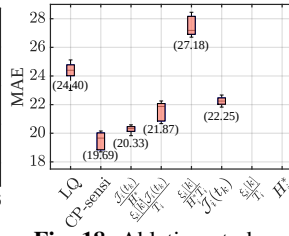


Fig. 18. Ablation study.

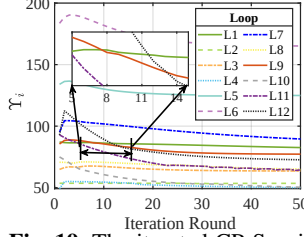


Fig. 19. The iterated CP-Sensi.

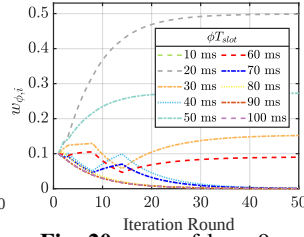


Fig. 20. $w_{\phi,i}$ of loop 9.

orities within each T_c . In the network with default scale, dissemination requires less than 0.4 s, making $T_c \leq 0.5$ s impractical. Conversely, when $T_c \geq 2.4$ s, control performance degrades as scheduling cannot respond quickly to frequently changing physical conditions.

C. Ablation Study

We evaluate the functionality of CP-Sensi's components by setting each of them as the scheduling priority, as shown in Fig. 18. $J_i(t_k)$ enhances control performance by 9.6% compared to LQ. Additionally, $\frac{J_i(t_k)}{H^*}$, which accounts for recourse usage and transmission conflicts, improves MAE by 17.1% over LQ. $\frac{J_i(t_k)}{T_i}$ integrates long- and short-term indicators but ignores the network factors. Ultimately, CP-Sensi achieves a 19.3% improvement in MAE over LQ.

D. Calibration of the Delay and Overhead Analyses

We present the iterated Υ_i and $w_{\phi,i}$ during the calibration of delay distribution estimations within one T_c , as shown in Figs. 19 and 20, respectively. We analyze the MAE and scheduling overhead for varying Ψ under the default network scale with $\Delta_i = 0.005$. In Fig. 19, each intersection of the two curves represents a priority swap, such as loops 1 and 9 after 8 iterations, and loops 9 and 12 after 13 iterations. Fig. 20 shows the iterated $w_{\phi,i}$ for different delay distributions ϕT_{slot} (legends in the figure), which adjusts accordingly after each priority swap. The MAE along with respect to Ψ is shown in Fig. 21. As Ψ increases, the MAE decreases, and the distribution becomes more concentrated, demonstrating the effectiveness of delay calibration. Each boxplot in Fig. 22 is based on 20,000 executions on a 2.6 GHz Intel Core i7 processor. Balancing MAE and overhead, we select a default $\Psi = 10$. The overhead of the scheduler remains below one T_s when $\Psi \leq 50$, illustrating the scalability of CP-Sensi.

E. A Joint Simulation over Lossy Networks

We evaluate CP-Sensi by integrating four double-tank systems implemented in Simulink, and an IEEE 802.15.4 mesh network implemented in TOSSIM. The simulated network

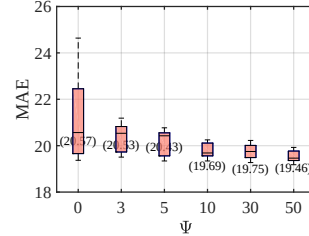


Fig. 21. The MAE with Ψ .

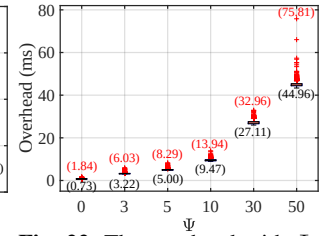


Fig. 22. The overhead with Ψ .

employs a topology of 16 TelosB nodes, with an average end-to-end distance of 4 hops from the testbed at Washington University [53]. The received signal strength indicator (RSSI) and noise strength of each link are -62 and -82 dBm, respectively, resulting in an average loss rate of 5%. Four double-tank models and their parameters are the same as those in [54], with $T_1 = 40$ ms, $T_2 = 40$ ms, $T_3 = 80$ ms, and $T_4 = 120$ ms. The simulation interval is 12 s. We present results for CP-Sensi and DPA, which performs the best among those baselines in Sec. VII-B.

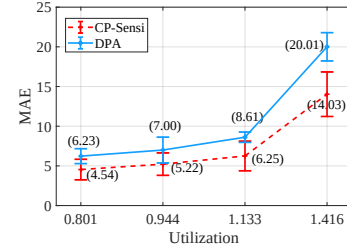


Fig. 23. MAEs of CP-Sensi and DPA in the joint simulation.

As shown in Fig. 23, CP-Sensi achieves a MAE improvement of up to 27.1% under $U < 1$ compared to DPA. When $U > 1$, the improvement becomes more significant, as CP-Sensi better captures the delay sensitivity, prioritizing the most sensitive flows. DPA increases the periods of flows to find a feasible schedule, which sacrifices the control performance. Moreover, the (mean, median, standard deviation) tuple of the scheduling overhead of DPA is (138.97, 135.65, 21.46) ms is much larger than (6.84, 6.70, 0.84) ms of CP-Sensi, which prevents it from timely response to dynamic physical states.

VIII. CONCLUSION

In this paper, we propose a mesh network scheduling based on cyber-physical sensitivity (CP-Sensi) for WCSs. To address challenges of increasing mesh network scale and the significant variation in scheduling latency across loops, we propose an arbitrary and fluctuating latency model to capture the impacts of delays on WCSs. We develop CP-Sensi, which integrates latency modeling, dynamic physical states, and network conditions to quantify sensitivity. The CP-Sensi-based mesh network scheduling combines the calibration of the delay distribution estimation with a reconfiguration protocol. Case studies involving 12 control loops under varying utilization levels and network scales demonstrate the significant improvement in control performance by CP-Sensi-based scheduling.

IX. ACKNOWLEDGEMENT

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REFERENCES

- [1] N. Tuptuk and S. Hailes, "Security of smart manufacturing systems," *Journal of manufacturing systems*, vol. 47, pp. 93–106, 2018.
- [2] G. Han, J. Tu, L. Liu, M. Martínez-García, and Y. Peng, "Anomaly detection based on multidimensional data processing for protecting vital devices in 6g-enabled massive IIoT," *IEEE Internet of Things Journal*, vol. 8, no. 7, pp. 5219–5229, 2021.
- [3] C. Lu, A. Saifullah, B. Li, M. Sha, H. Gonzalez, D. Gunatilaka, C. Wu, L. Nie, and Y. Chen, "Real-time wireless sensor-actuator networks for industrial cyber-physical systems," *Proceedings of the IEEE*, vol. 104, no. 5, pp. 1013–1024, 2016.
- [4] H. Zhang, Z. Zhang, Z. Wang, and Q. Shan, "New results on stability and stabilization of networked control systems with short time-varying delay," *IEEE transactions on cybernetics*, vol. 46, no. 12, pp. 2772–2781, 2015.
- [5] H. Luo and Z. Hu, "Stability analysis of sampled-data load frequency control systems with multiple delays," *IEEE Transactions on Control Systems Technology*, vol. 30, no. 1, pp. 434–442, 2021.
- [6] F. Mager, D. Baumann, C. Herrmann, S. Trimpe, and M. Zimmerling, "Scaling beyond bandwidth limitations: Wireless control with stability guarantees under overload," *ACM Transactions on Cyber-Physical Systems (TCPS)*, vol. 6, no. 3, pp. 1–30, 2022.
- [7] S. Ghosh, S. Dey, and P. Dasgupta, "Pattern guided integrated scheduling and routing in multi-hop control networks," *ACM Transactions on Embedded Computing Systems (TECS)*, vol. 19, no. 2, pp. 1–28, 2020.
- [8] M. Doostmohammadian, U. A. Khan, M. Pirani, and T. Charalambous, "Consensus-based distributed estimation in the presence of heterogeneous, time-invariant delays," *IEEE Control Systems Letters*, vol. 6, pp. 1598–1603, 2021.
- [9] J. Wang, R. De Leone, S. Fu, J. Xia, and L. Qiao, "Event-triggered control design for networked evolutionary games with time invariant delay in strategies," *International Journal of Systems Science*, vol. 52, no. 3, pp. 493–504, 2021.
- [10] L. Gu, D. Zeng, S. Tao, S. Guo, H. Jin, A. Y. Zomaya, and W. Zhuang, "Fairness-aware dynamic rate control and flow scheduling for network utility maximization in network service chain," *IEEE Journal on Selected Areas in Communications*, vol. 37, no. 5, pp. 1059–1071, 2019.
- [11] M. J. Neely, "Delay-based network utility maximization," *IEEE/ACM Transactions on Networking*, vol. 21, no. 1, pp. 41–54, 2012.
- [12] M. H. Mamduhi, D. Maity, S. Hirche, J. S. Baras, and K. H. Johansson, "Delay-sensitive joint optimal control and resource management in multiloop networked control systems," *IEEE Transactions on Control of Network Systems*, vol. 8, no. 3, pp. 1093–1106, 2021.
- [13] H. S. Chwa, K. G. Shin, and J. Lee, "Closing the gap between stability and schedulability: A new task model for cyber-physical systems," in *2018 IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS)*. IEEE, 2018, pp. 327–337.
- [14] W. Wang, D. Mosse, and A. V. Papadopoulos, "Packet priority assignment for wireless control systems of multiple physical systems," *Journal of Systems Architecture*, vol. 107, p. 101708, 2020.
- [15] G. Chen, X. Cao, L. Liu, C. Sun, and Y. Cheng, "Joint scheduling and channel allocation for end-to-end delay minimization in industrial wireless networks," *IEEE Internet of Things Journal*, vol. 6, no. 2, pp. 2829–2842, 2018.
- [16] J. J. G. García and M. G. Harbour, "Minimizing the effects of jitter in distributed hard real-time systems," *Journal of systems architecture*, vol. 42, no. 6–7, pp. 431–447, 1996.
- [17] P. Park, S. C. Ergen, C. Fischione, C. Lu, and K. H. Johansson, "Wireless network design for control systems: A survey," *IEEE Communications Surveys & Tutorials*, vol. 20, no. 2, pp. 978–1013, 2017.
- [18] M. Klügel, M. Mamduhi, O. Ayan, M. Vilgelm, K. H. Johansson, S. Hirche, and W. Kellerer, "Joint cross-layer optimization in real-time networked control systems," *IEEE Transactions on Control of Network Systems*, vol. 7, no. 4, pp. 1903–1915, 2020.
- [19] G. Chen, X. Cao, and J. Jin, "Joint scheduling and channel allocation for kalman filtering over multihop wireless networks," *IEEE Transactions on Industrial Informatics*, vol. 17, no. 5, pp. 3555–3565, 2020.
- [20] A. Jain, P. Modekurthy, and A. Saifullah, "Control over low-power wide-area networks," in *2024 ACM/IEEE 15th International Conference on Cyber-Physical Systems (ICCPs)*. IEEE, 2024, pp. 192–201.
- [21] L. Chen, B. Hu, Z.-H. Guan, L. Zhao, and D.-X. Zhang, "Control-aware transmission scheduling for industrial network systems over a shared communication medium," *IEEE Internet of Things Journal*, vol. 9, no. 13, pp. 11 299–11 310, 2021.
- [22] D. Roy, S. Ghosh, Q. Zhu, M. Caccamo, and S. Chakraborty, "Good-spread: Criticality-aware static scheduling of cps with multi-qos resources," in *2020 IEEE Real-Time Systems Symposium (RTSS)*. IEEE, 2020, pp. 178–190.
- [23] O. Ayan, S. Hirche, A. Ephremides, and W. Kellerer, "Optimal finite horizon scheduling of wireless networked control systems," *IEEE/ACM Transactions on Networking*, 2023.
- [24] F. Ferrari, M. Zimmerling, L. Mottola, and L. Thiele, "Low-power wireless bus," in *Proceedings of the 10th ACM Conference on Embedded Network Sensor Systems (SenSys)*, 2012, pp. 1–14.
- [25] F. Ferrari, M. Zimmerling, L. Thiele, and O. Saukh, "Efficient network flooding and time synchronization with glossy," in *Proceedings of the 10th ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN)*. IEEE, 2011, pp. 73–84.
- [26] V. P. Modekurthy and A. Saifullah, "Online period selection for wireless control systems," in *2019 IEEE International Conference on Industrial Internet (ICII)*. IEEE, 2019, pp. 170–179.
- [27] Y. Ma, C. Lu, and Y. Wang, "Efficient holistic control: Self-awareness across controllers and wireless networks," *ACM Transactions on Cyber-Physical Systems*, vol. 4, no. 4, pp. 1–27, 2020.
- [28] S. Kaul, R. Yates, and M. Gruteser, "On piggybacking in vehicular networks," in *2011 IEEE Global Telecommunications Conference-GLOBECOM 2011*. IEEE, 2011, pp. 1–5.
- [29] L. An and G.-H. Yang, "Data-based distributed sensor scheduling for multiple linear systems with H_∞ performance preservation," *IEEE Transactions on Automatic Control*, vol. 67, no. 12, 2021.
- [30] D. Soudbakhsh, A. Annaswamy, and H. Voit, "Adaptation in network control systems with hierarchical scheduling," *IET Control Theory & Applications*, vol. 13, no. 17, pp. 2775–2782, 2019.
- [31] WirelessHART. [Online]. Available: <http://en.hartcomm.org/>
- [32] H. Yu, P. Zeng, and C. Xu, "Industrial wireless control networks: From wia to the future," *Engineering*, vol. 8, pp. 18–24, 2022.
- [33] Q. Leng, Y.-H. Wei, S. Han, A. K. Mok, W. Zhang, and M. Tomizuka, "Improving control performance by minimizing jitter in rt-wifi networks," in *2014 IEEE Real-Time Systems Symposium (RTSS)*. IEEE, 2014, pp. 63–73.
- [34] T. Zhang, T. Gong, S. Han, Q. Deng, and X. S. Hu, "Distributed dynamic packet scheduling framework for handling disturbances in real-time wireless networks," *IEEE Transactions on Mobile Computing*, vol. 18, no. 11, pp. 2502–2517, 2018.
- [35] T. Zhang, T. Gong, C. Gu, H. Ji, S. Han, Q. Deng, and X. S. Hu, "Distributed dynamic packet scheduling for handling disturbances in real-time wireless networks," in *2017 IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS)*. IEEE, 2017, pp. 261–272.
- [36] S. Hong, X. S. Hu, T. Gong, and S. Han, "On-line data link layer scheduling in wireless networked control systems," in *2015 27th Euromicro Conference on Real-Time Systems (ECRTS)*. IEEE, 2015, pp. 57–66.
- [37] Y. Ma and C. Lu, "Efficient holistic control over industrial wireless sensor-actuator networks," in *2018 IEEE International Conference on Industrial Internet (ICII)*. IEEE, 2018, pp. 89–98.
- [38] B. Li, L. Nie, C. Wu, H. Gonzalez, and C. Lu, "Incorporating emergency alarms in reliable wireless process control," in *Proceedings of the ACM/IEEE Sixth International Conference on Cyber-Physical Systems (ICCPs)*, 2015, pp. 218–227.
- [39] B. Li, Y. Ma, T. Westenbroek, C. Wu, H. Gonzalez, and C. Lu, "Wireless routing and control: A cyber-physical case study," in *2016 ACM/IEEE 7th International Conference on Cyber-Physical Systems (ICCPs)*. IEEE, 2016, pp. 1–10.
- [40] Q. Ding, C. Luo, H. Yang, and Y. Luo, "Throughput maximization of dynamic tdd networks with a full-duplex uav-bs," *IEEE Transactions on Wireless Communications*, 2024.
- [41] H. Lin and P. J. Antsaklis, "Stability and stabilizability of switched linear systems: a survey of recent results," *IEEE Transactions on Automatic control*, vol. 54, no. 2, pp. 308–322, 2009.
- [42] D. Baumann, F. Mager, R. Jacob, L. Thiele, M. Zimmerling, and S. Trimpe, "Fast feedback control over multi-hop wireless networks with mode changes and stability guarantees," *ACM Transactions on Cyber-Physical Systems*, vol. 4, no. 2, pp. 1–32, 2019.
- [43] K. Gu, J. Chen, and V. L. Kharitonov, *Stability of time-delay systems*. Springer Science & Business Media, 2003.

- [44] N. S. Nise, *Control systems engineering*. John Wiley & Sons, 2020.
- [45] A. Saifullah, Y. Xu, C. Lu, and Y. Chen, "End-to-end delay analysis for fixed priority scheduling in wireless networks," in *2011 17th IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS)*. IEEE, 2011, pp. 13–22.
- [46] V. P. Modekurthy, M. Rahman, and A. Saifullah, "Towards mixed criticality industrial wireless sensor-actuator network," in *Proceedings of the 24th International Conference on Distributed Computing and Networking (ICDCN)*, 2023, pp. 425–430.
- [47] G. C. Buttazzo and G. Buttazzo, *Hard real-time computing systems*. Springer, 1997, vol. 356.
- [48] E. Bini and G. C. Buttazzo, "Schedulability analysis of periodic fixed priority systems," *IEEE Transactions on Computers*, vol. 53, no. 11, pp. 1462–1473, 2004.
- [49] R. Gifford, F. Galarza-Jimenez, L. T. X. Phan, and M. Zamani, "Decntr: Optimizing safety and schedulability with multi-mode control and resource allocation co-design," in *2024 IEEE 30th Real-Time and Embedded Technology and Applications Symposium (RTAS)*. IEEE, 2024, pp. 306–319.
- [50] A. Derhab, F. Ounini, and B. Remli, "Mob-tossim: An extension framework for tossim simulator to support mobility in wireless sensor and actuator networks," in *2012 IEEE 8th International Conference on Distributed Computing in Sensor Systems (DCOSS)*. IEEE, 2012, pp. 300–305.
- [51] M. A. Haque, D. Ismail, and A. Saifullah, "Scalable real-time control in industrial cyber-physical systems," in *Proceedings of the 25th International Conference on Distributed Computing and Networking (ICDCN)*, 2024, pp. 353–358.
- [52] A. Saifullah, C. Wu, P. B. Tiwari, Y. Xu, Y. Fu, C. Lu, and Y. Chen, "Near optimal rate selection for wireless control systems," *ACM Transactions on Embedded Computing Systems (TECS)*, vol. 13, no. 4s, pp. 1–25, 2014.
- [53] (2025) The wustl wireless sensor network testbed. [Online]. Available: https://wsn.cse.wustl.edu/index.php?title=The_WUSTL_Wireless_Sensor_Network_Testbed
- [54] Y. Ma, J. Guo, Y. Wang, A. Chakrabarty, H. Ahn, P. Orlik, X. Guan, and C. Lu, "Optimal dynamic transmission scheduling for wireless networked control systems," *IEEE Transactions on Control Systems Technology*, vol. 30, no. 6, pp. 2360–2376, 2022.